

宁波大学
NINGBO UNIVERSITY

实 验 报 告

实验二 ZigBee 无线传感器组网实践

姓名：	顾凯琪
学号：	226001624
课程名称：	网络工程课程设计
所在学院：	信息科学与工程学院
专业班级：	2022 计算机科学与技术 2 班
任课教师：	江先亮

实验项目名称	**实验二 ZigBee 无线传感器组网实践**

一、 实验目的与要求：

1)了解 ZigBee 网络基本应用

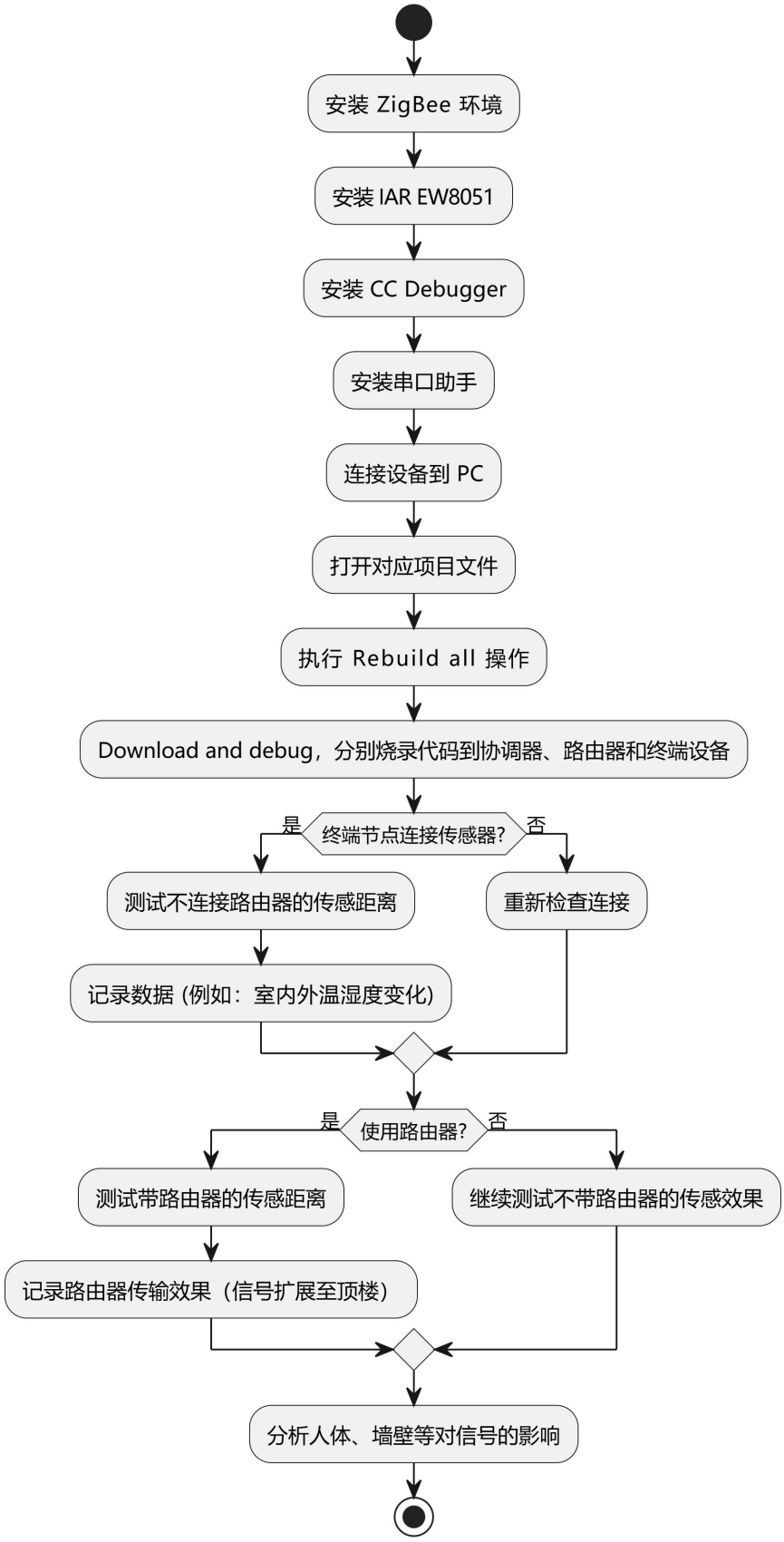
2)掌握 ZigBee 节点结构和组建

3)熟悉 ZigBee 户外环境监测

二、 实验环境：

笔记本 PC 一台、CC2530 传感器节点套件（CC2530 节点、CC Debugger 调试器、USB A 型转 B 型线、下载线、Mini USB 线、天线）、电池、Z-Stack 协议栈、IAR Embedded Workbench 软件、Smart Flash Programmer 软件、ZigBee_Packet_Sniffer 软件、ZigBee_Sensor_Monitor 软件、温湿度传感器若干、电池及 5V 稳压模块若干、母对母杜邦线若干、SSCOM 串口调试助手软件、CC2530 节点代码。

三、　实验方法（原理、流程图）


安装 ZigBee 环境
安装 IAR EW8051
安装 CC Debugger
安装串口助手
连接设备到 PC
打开对应项目文件
执行 Rebuild all 操作
Download and debug，分别烧录代码到协调器、路由器和终端设备
是
终端节点连接传感器?
否
测试不连接路由器的传感距离
重新检查连接
记录数据 (例如：室内外温湿度变化)
是
使用路由器?
否
测试带路由器的传感距离
继续测试不带路由器的传感效果
记录路由器传输效果（信号扩展至顶楼）
分析人体、墙壁等对信号的影响

四、　实验过程、步骤及内容

安装 ZigBee



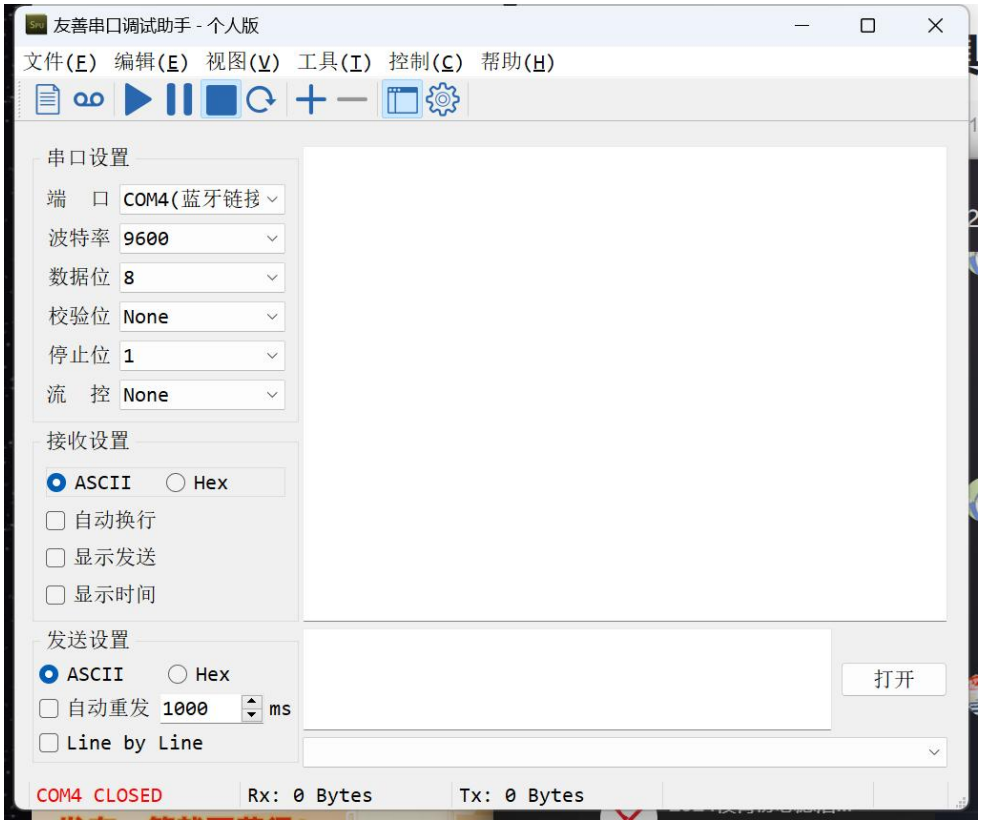
安装 IAR EW8051



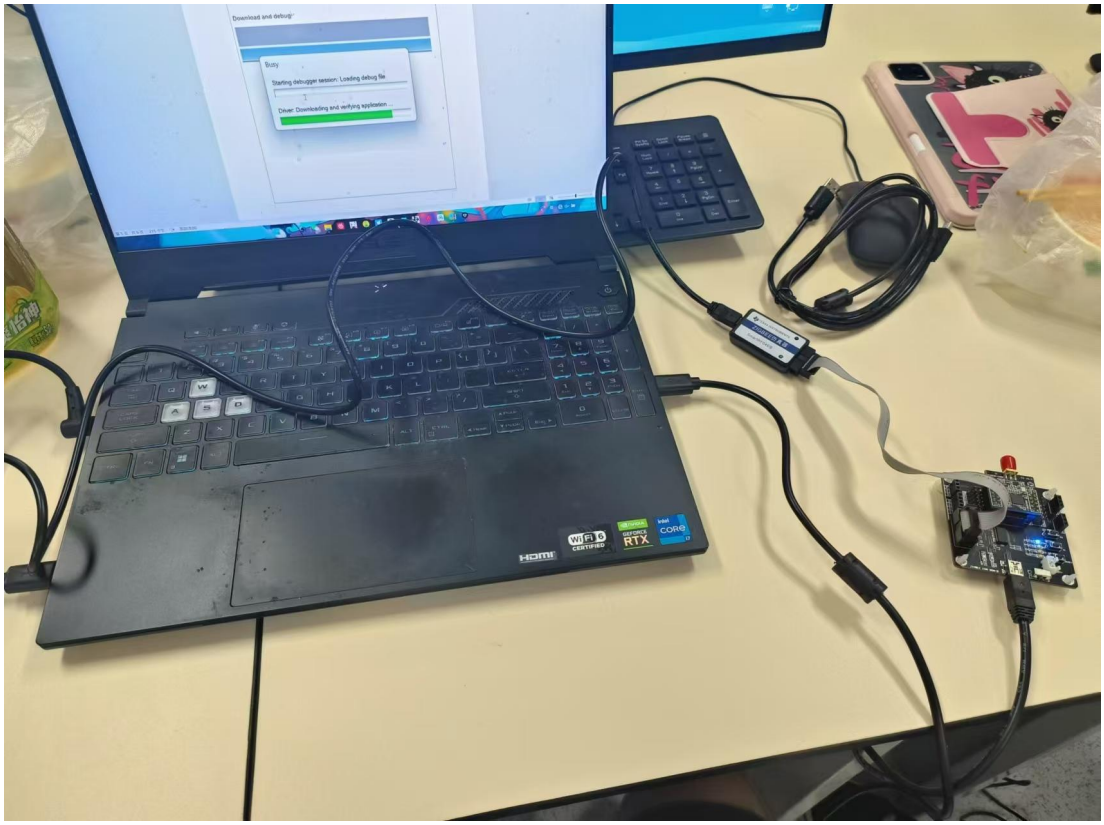
安装 CC Debugger



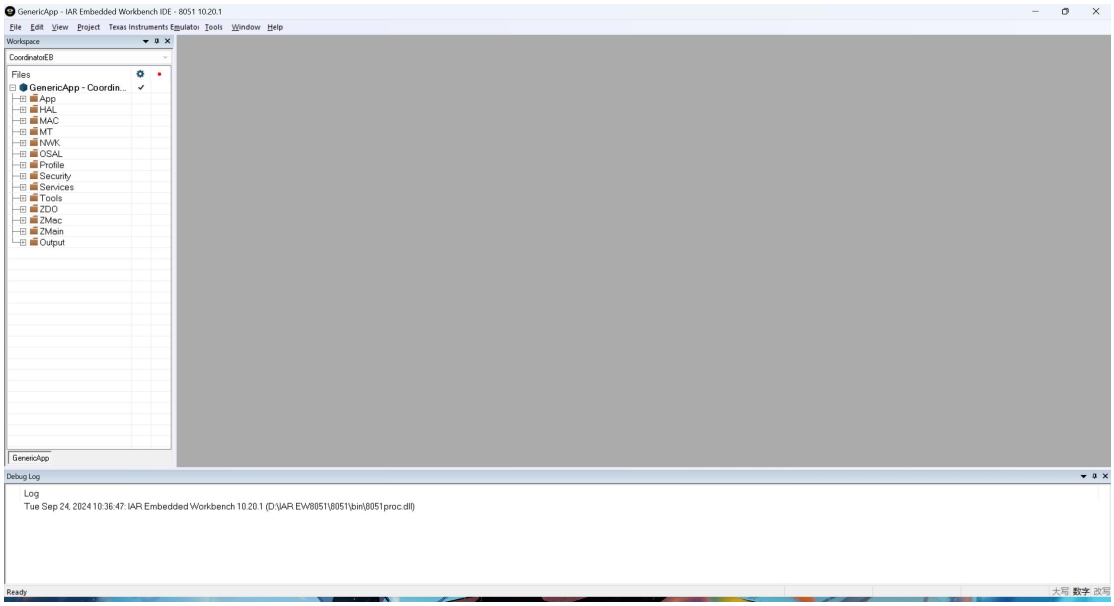
安装串口助手



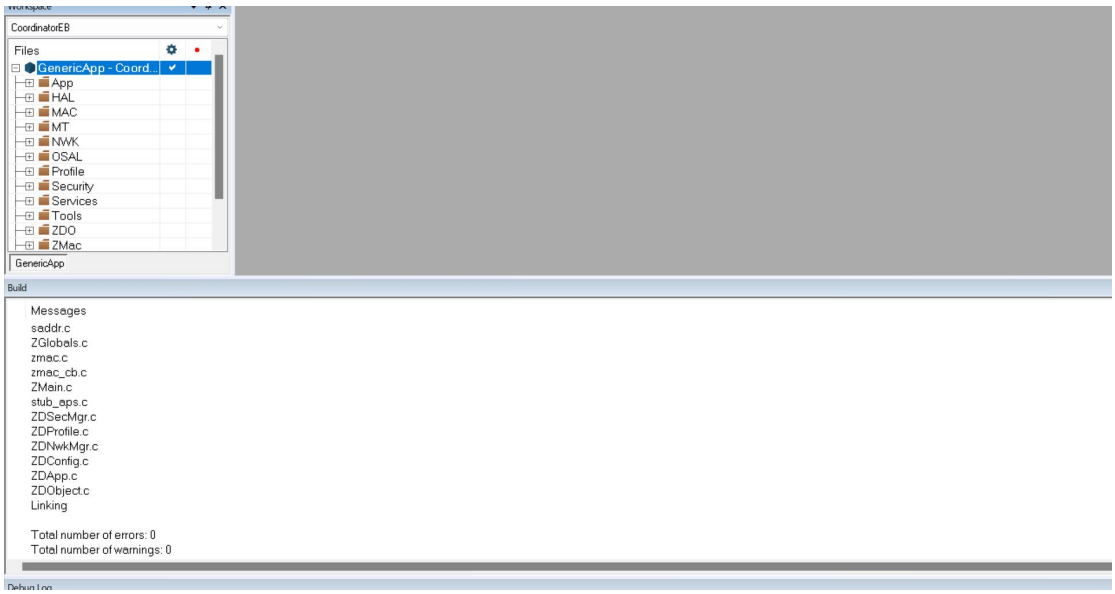
将对应设备连接 PC，等待烧录



打开对应项目文件

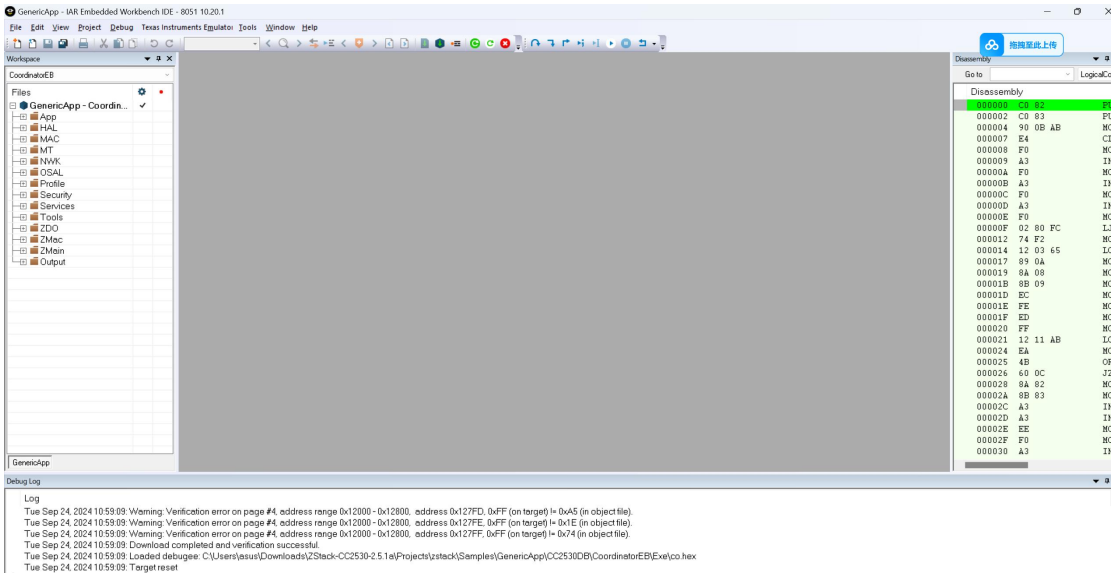


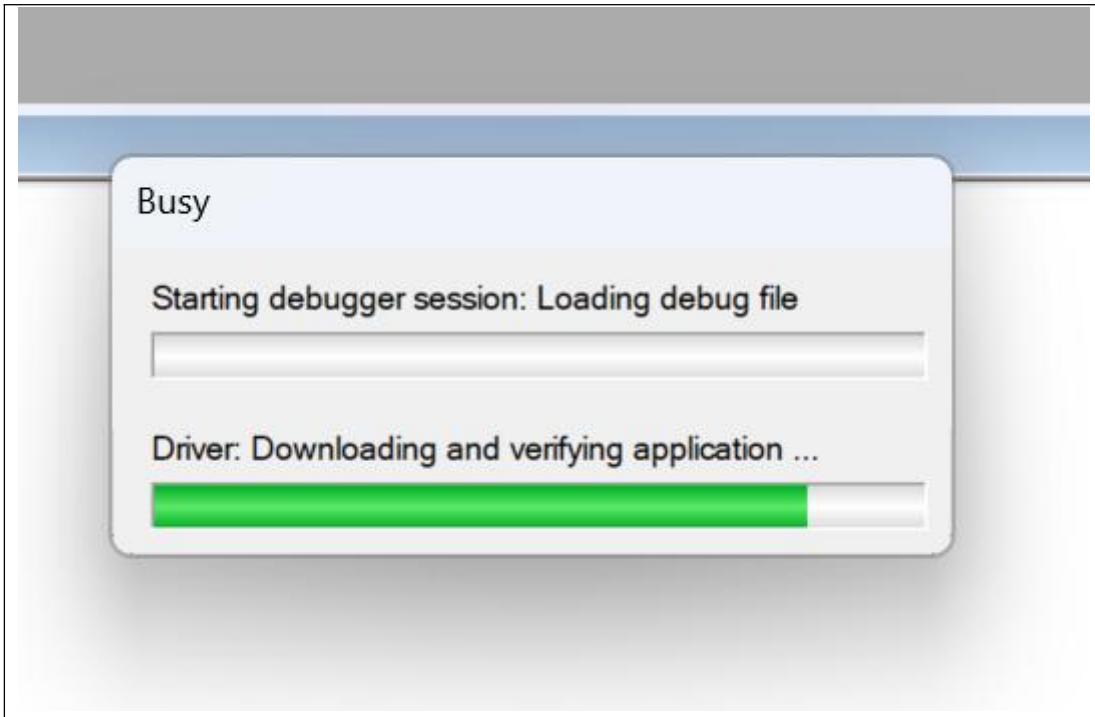
执行 Rebuild all 操作，为 2530 节点重置代码



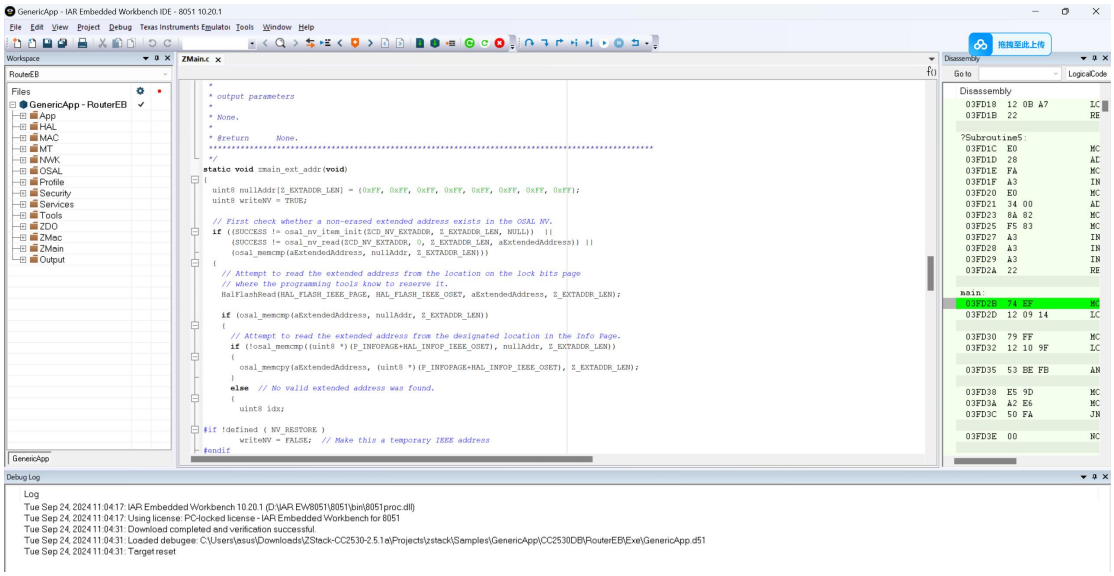
Download and debug 将对应不同设备（协调器，路由器，终端）的代码烧录进对应的 2530 节点

协调器

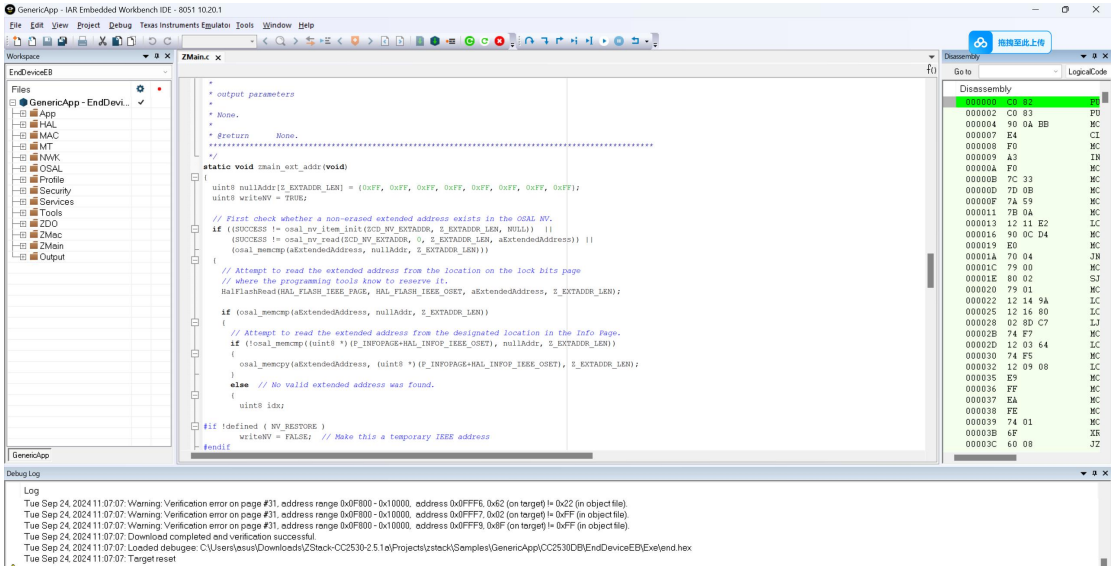



Busy
Starting debugger session: Loading debug file
Driver: Downloading and verifying application ...

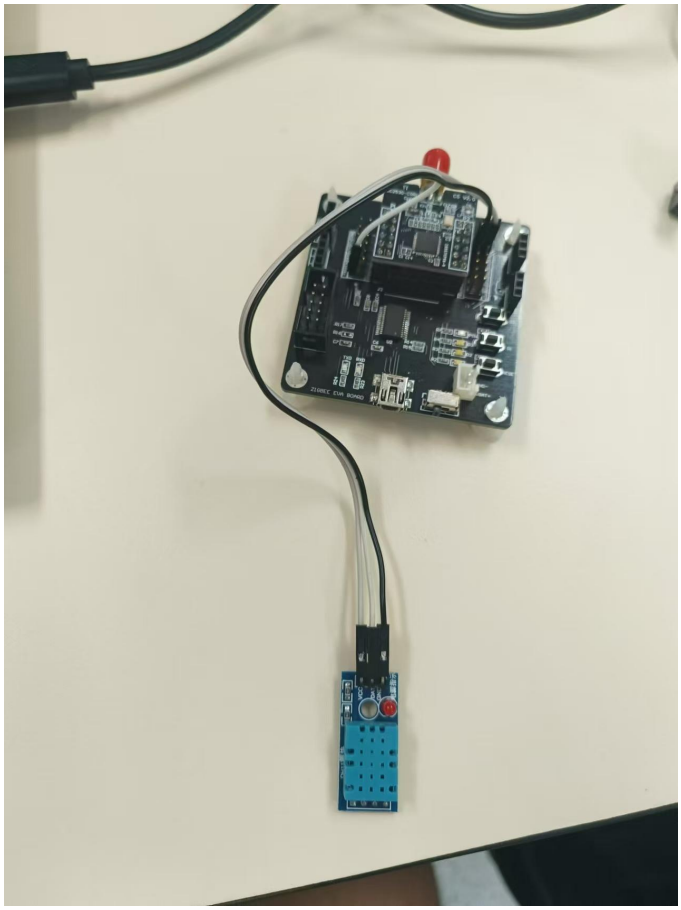
路由器


GenericApp - IAR Embedded Workbench IDE - 8051 10.20.1
File Edit View Project Debug Texas Instruments Emulator Tools Window Help
Workspace
RouterEB
Files
GenericApp - RouterEB
App
HAL
MAC
MT
NWK
OSAL
Profile
Security
Services
Tools
ZDO
ZMac
ZMain
Output
GenericApp
ZMain.c
拖拽至此上传
Disassembly
Go to
LogicalCode
Debug Log
Log
Tue Sep 24, 2024 11:04:17: IAR Embedded Workbench 10.20.1 (D:\IAR EW8051\8051\bin\8051proc.dll)
Tue Sep 24, 2024 11:04:17: Using license: PC-locked license - IAR Embedded Workbench for 8051
Tue Sep 24, 2024 11:04:31: Download completed and verification successful.
Tue Sep 24, 2024 11:04:31: Loaded debugee: C:\Users\asus\Downloads\ZStack-CC2530-2.5.1a\Projects\zstack\Samples\GenericApp\CC2530DB\RouterEB\Exe\GenericApp.d51
Tue Sep 24, 2024 11:04:31: Target reset

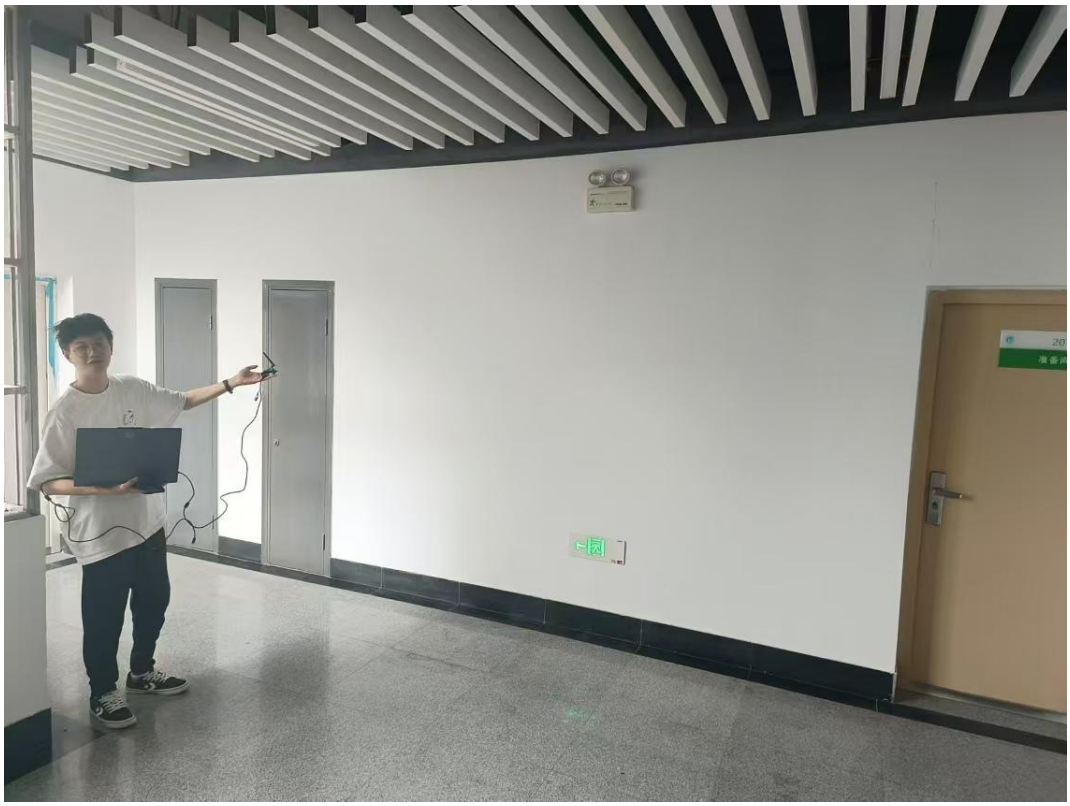
终端

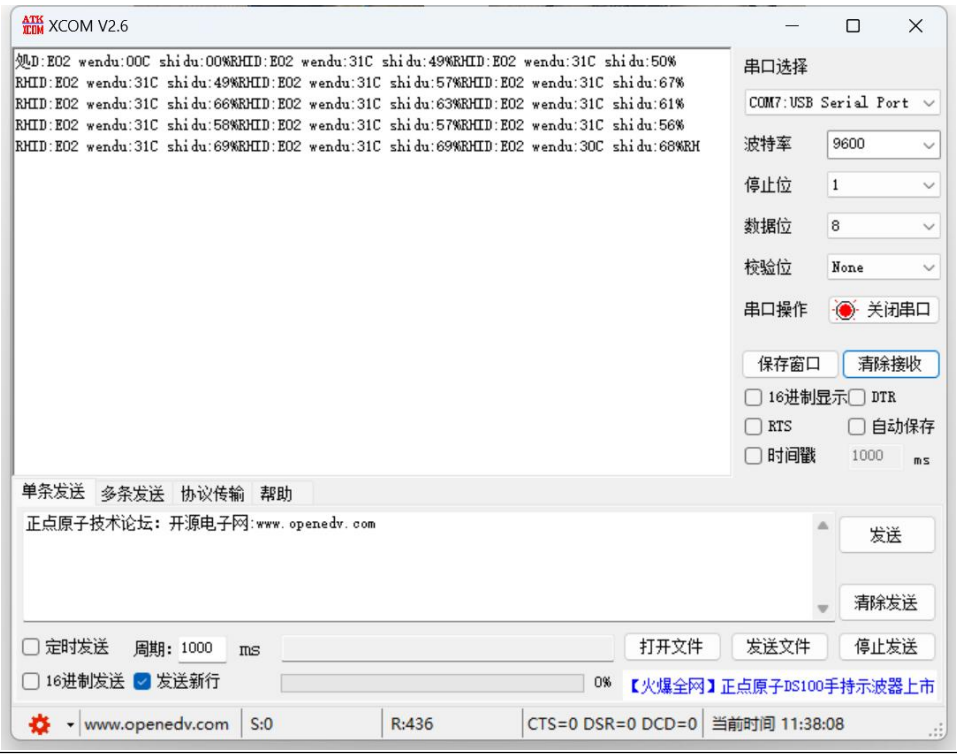

GenericApp - IAR Embedded Workbench IDE - 8051 10.20.1
File Edit View Project Debug Texas Instruments Emulator Tools Window Help
Workspace
EndDeviceEB
Files
GenericApp - EndDevi...
App
HAL
MAC
MT
NWK
OSAL
Profile
Security
Services
Tools
ZDO
ZMac
ZMain
Output
GenericApp
ZMain.c
static void zmain_ext_addr(void)
{
uint8 nullAddr[Z_EXTADDR_LEN] = {0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF};
uint8 writeNV = TRUE;
// First check whether a non-erased extended address exists in the OSAL NV.
if ((SUCCESS != osal_nv_item_init(ZCD_NV_EXTADDR, Z_EXTADDR_LEN, NULL)) ||
(SUCCESS != osal_nv_read(ZCD_NV_EXTADDR, 0, Z_EXTADDR_LEN, aExtendedAddress)) ||
(osal_memcmp(aExtendedAddress, nullAddr, Z_EXTADDR_LEN)))
{
// Attempt to read the extended address from the location on the lock bits page
// where the programming tools know to reserve it.
HalFlashRead(HAL_FLASH_IEEE_PAGE, HAL_FLASH_IEEE_OSET, aExtendedAddress, Z_EXTADDR_LEN);
if (osal_memcmp(aExtendedAddress, nullAddr, Z_EXTADDR_LEN))
{
// Attempt to read the extended address from the designated location in the Info Page.
if (!osal_memcmp((uint8 *)(P_INFOPAGE+HAL_INFOP_IEEE_OSET), nullAddr, Z_EXTADDR_LEN))
{
osal_memcpy(aExtendedAddress, (uint8 *)(P_INFOPAGE+HAL_INFOP_IEEE_OSET), Z_EXTADDR_LEN);
}
else // No valid extended address was found.
{
uint8 idx;
#if !defined (NV_RESTORE)
writeNV = FALSE; // Make this a temporary IEEE address
#endif
Disassembly
Go to
LogicalCode
000001 C0 82 PU
000002 C0 83 PU
000004 90 0A BB MC
000007 E4 CI
000008 F0 MC
000009 A3 IN
00000A F0 MC
00000B 7C 33 MC
00000D 7D 0B MC
00000F 7A 59 MC
000011 7B 0A MC
000013 12 11 E2 LC
000016 90 0C D4 MC
000019 E0 MC
00001A 70 04 JN
00001C 79 00 MC
00001E 80 02 SJ
000020 79 01 MC
000022 12 14 9A LC
000025 12 16 80 LC
000028 02 8D C7 LJ
00002B 74 F7 MC
00002D 12 03 64 LC
000030 74 F5 MC
000032 12 09 08 LC
000035 E9 MC
000036 FF MC
000037 EA MC
000038 FE MC
000039 74 01 MC
00003B 6F XR
00003C 60 08 JZ
Debug Log
Log
Tue Sep 24, 2024 11:07:07: Warning: Verification error on page #31, address range 0x0F800 - 0x10000, address 0x0FFF6, 0x62 (on target) != 0x22 (in object file).
Tue Sep 24, 2024 11:07:07: Warning: Verification error on page #31, address range 0x0F800 - 0x10000, address 0x0FFF7, 0x02 (on target) != 0xFF (in object file).
Tue Sep 24, 2024 11:07:07: Warning: Verification error on page #31, address range 0x0F800 - 0x10000, address 0x0FFF9, 0x8F (on target) != 0xFF (in object file).
Tue Sep 24, 2024 11:07:07: Download completed and verification successful.
Tue Sep 24, 2024 11:07:07: Loaded debugee: C:\Users\asus\Downloads\ZStack-CC2530-2.5.1a\Projects\zstack\Samples\GenericApp\CC2530DB\EndDeviceEB\Exe\end.hex
Tue Sep 24, 2024 11:07:07: Target reset

将终端连接传感器

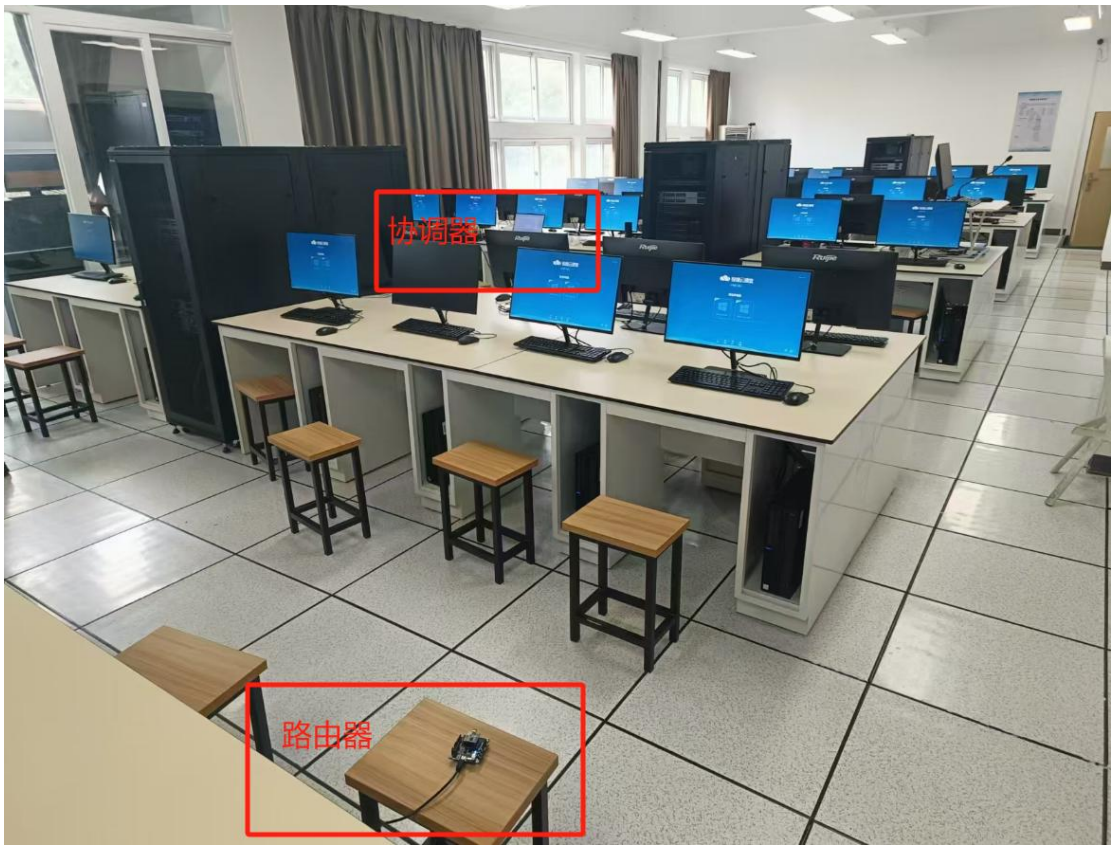


首先尝试不连接路由器，测试无限传感距离，如图所示，传感的传输距离一直扩展到两堵墙之外的走廊，才接受不到信号，具体的传感信号如下所示（可以观察到湿度从室内到室外有个明显的变化）





之后测试带有路由器作为中间节点，如下所示安置路由器和协调器，本次实验，另一位同学一直走到顶楼信号也未断绝，此路由器效果有点过于强大



另外，经过测试，人体的阻挡对拥有天线的节点而言影响微乎其微，墙壁的阻挡远大于空间距离的影响

五、 实验结论

本次实验验证了 ZigBee 无线传感器网络在环境监测中的应用，通过协调器、路由器和终端节点的组合，测试了传输距离和信号强度。实验结果表明，路由器有效扩展了传输范围，墙壁阻挡对信号影响较大，而人体阻挡影响较小，ZigBee 网络在户外环境中具有较好的稳定性和抗干扰能力。同时路由器有效的扩大了信号的传输范围。

六、 教师批阅意见：

成绩评定：

教师签字：

年 月 日

七、 备注：